METRIC

MIL-DTL-87173D
w/AMENDMENT 1
6 May 2024

SUPERSEDING
MIL-DTL-87173D
12 July 2023

DETAIL SPECIFICATION

PROPELLANT, PRIMING FLUID ALCM, GRADE PF-1

This Specification is Inactive for new design as of 13 February 1998.

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of high density synthetic hydrocarbon type priming fluid.

1.2 Classification.

Grade	Description
PF-1	Priming Fluid composed solely of exo-tetrahydrodicyclopentadiene and methylcyclohexane

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9135

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 and section 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 and section 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-85470 - Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-290 - Packing and Marking of Petroleum and Related Products

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D56 - Standard Test Method for Flash Point by Tag Closed Cup Tester

ASTM D93 - Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester

ASTM D156 - Standard Test Method for Saybolt Color of Petroleum Products (Saybolt Chromometer Method)

ASTM D240 - Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter

ASTM D381 - Standard Test Method for Gum Content in Fuels by Jet Evaporation

ASTM D445 - Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)

ASTM D1298 - Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method

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ASTM D2276 - Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling

ASTM D3241 - Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels

ASTM D3828 - Standard Test Methods for Flash Point by Small Scale Closed Cup Tester

ASTM D4052 - Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D4177 - Standard Practice for Automatic Sampling of Petroleum and Petroleum Products

ASTM D4306 - Standard Practice for Aviation Fuel Sample Containers for Tests Affected by Trace Contamination

ASTM D4809 - Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (Precision Method)

ASTM D5006 - Standard Test Method for Measurement of Fuel System Icing Inhibitors (Ether Type) in Aviation Fuels

ASTM D5452 - Standard Test Method for Particulate Contamination in Aviation Fuels by Laboratory Filtration

ASTM D6045 - Standard Test Method for Color of Petroleum Products by the Automatic Tristimulus Method

ASTM D6986 - Standard Test Method for Free Water, Particulate and Other Contamination in Aviation Fuels (Visual Inspection Procedures)

ASTM D7042 - Standard Test Method for Dynamic Viscosity and Density of Liquids by Stabinger Viscometer (and the Calculation of Kinematic Viscosity)

ASTM D7236 - Standard Test Method for Flash Point by Small Scale Closed Cup Tester (Ramp Method)

ASTM D7619 - Standard Test Method for Sizing and Counting Particles in Light and Middle Distillate Fuels, by Automatic Particle Counter

ASTM D7777 - Standard Test Method for Density, Relative Density, or API Gravity of Liquid Petroleum by Portable Digital Density Meter

ASTM D7945 - Standard Test Method for Determination of Dynamic Viscosity and Derived Kinematic Viscosity of Liquids by Constant Pressure Viscometer

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ASTM E29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available from https://www.astm.org.)

ENERGY INSTITUTE (EI)

IP 170 - Determination of flash point – Abel closed-cup method

IP 540 - Determination of the existent gum content of aviation fuel – Jet evaporation method

IP 565 - Determination of the level of cleanliness of aviation turbine fuel – Portable automatic particle counter method

IP 577 - Determination of the level of cleanliness of aviation turbine fuel – Automatic particle counter method using light extinction

(Copies of these documents are available from https://publishing.energyinst.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4406 - Hydraulic fluid power — Fluids — Method for coding the level of contamination by solid particles

ISO 11171 - Hydraulic Fluid Power – Calibration of Automatic Particle Counters for Liquids

(Copies of these documents are available from https://www.iso.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Except as otherwise specified herein, the priming fluid shall consist solely of exo-tetrahydrodicyclopentadiene and methylcyclohexane.

3.1.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Finished priming fluid. Finished priming fluid shall meet the requirements of this specification.

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3.3 Additives. The type and amount of each additive used shall be made available when requested by procuring activity or user (see 6.2.d). The only additives approved for use are those referenced in this specification.

3.3.1 Antioxidants. Immediately after processing and before the propellant is exposed to the atmosphere, add an approved antioxidant formulation (3.3.1.1) or combination of approved antioxidant formulations in order to prevent the formation of polymeric oxidation products. The concentration of antioxidant to be added shall not be less than 81.0 milligrams (mg) nor more than 102.3 mg of active ingredient per liter of priming fluid (90 parts per million (min) to 110 parts per million (max) by weight, not including weight of solvent).

3.3.1.1 Antioxidant formulations. The following antioxidant formulations are approved:

a. 2,6-di-tert-butyl-4-methyl-phenol

b. 2,4-dimethyl-6-tert-butyl-phenol

c. 2,6-di-tert-butyl-phenol

3.3.2 Fuel system icing inhibitor. Fuel system icing inhibitor (FSII) shall only be added at the direction of the acquiring official. When required, the FSII shall conform to MIL-DTL-85470.

3.3.3 Additive mixing. To prevent any possible reaction between the concentrated forms of the different additives, the additives shall not be commingled prior to their addition to the priming fluid.

3.4 Workmanship. At the time of Government acceptance, the finished priming fluid shall be visually free from undissolved water, sediment, or suspended matter. Refer to ASTM D6986. In case of dispute, the priming fluid shall be clear and bright at 21 °C (70 °F) and meet the Table I particulate matter requirement.

TABLE I. Chemical and physical requirements and test methods.

Property	Minimum	Maximum	Test Method or Test Paragraph
Workmanship			3.4
Color, Saybolt	+25		D156 [1], D6045

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TABLE I. Chemical and physical requirements and test methods - Continued.

Property	Minimum	Maximum	Test Method or Test Paragraph
Chemical Analysis, weight percent			Appendix A
exo-tetrahydrodicyclopentadiene [2]	83.5	90.0	
Methylcyclohexane [2]	10.0	15.0	
other hydrocarbons [2]		1.5	
Flash point, °C (°F)	16(60)	27(80)	D93 [1], D56 [3], D3828 [3], D7236 [3], IP170 [3]
Density, 15 °C, kg/L (API Gravity)	0.900 (25.7)	0.930 (20.7)	D1298, D4052 [1], D7777
Viscosity, @-54°C (-65°F), mm^2/s		40	D445 [1], D7042 [4], D7945
Net heat of combustion			D240, D4809 [1]
MJ/kg (BTU/lb)	42.1 (18,100)		
MJ/m^3 (BTU/gal)	37,890 (135,900)		
FSII, vol. percent [5]	0.10	0.15	D5006
Existent gum, mg/100 mL		5	IP 540, D381 [1,6]
Thermal Stability (2.5 hr at 300°C (572°F))			D3241
Change in pressure drop, mm Hg		10	
Tube rating: One of the following shall be measured.			
(1) Annex A1 VTR		<3 [7]	
(2) Annex A2, ITR [8], nm		50	
(3) Annex A3, ETR [1], nm		50	
Particulate Matter [9]			
Gravimetric, mg/L		1.0	D5452 [1], D2276
Particle Counting [10], ISO Code		19/17/14/13	IP 565, IP 577, D7619 [1]

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TABLE I. Chemical and physical requirements and test methods - Continued.

<table>
<tr><th colspan="2">NOTES:</th></tr>
<tr><td>1.</td><td>Referee test method</td></tr>
<tr><td>2.</td><td>Test procedure and required equipment outlined in Appendix A.</td></tr>
<tr><td>3.</td><td>Relative to ASTM D93: results obtained by D56 can be up to 1.5°C lower, results obtained by D3828 can be up to 2°C lower, and results obtained by D7236 and IP 170 can be up to 1.0 °C lower.</td></tr>
<tr><td>4.</td><td>ASTM D7042 results shall be corrected to bias-free viscosity results by the application of the correction factor in the D7042 Precision and Bias section.</td></tr>
<tr><td>5.</td><td>As required by acquiring official, see 3.3.2.</td></tr>
<tr><td>6.</td><td>The preferred vaporizing medium for the priming fluid is steam; however, the existent gum test IP 540 may be performed using air as the vaporizing medium. If air is used instead of steam, it shall be recorded. Test Method ASTM D381, using steam jet operating conditions, shall be the referee test method.</td></tr>
<tr><td>7.</td><td>Peacock (code P) or Abnormal color deposits (code A) result in a failure.</td></tr>
<tr><td>8</td><td>If the ITR reports "N/A" for the tube's volume measurement, the result constitutes a failure and the result shall be reported as ">50 nm."</td></tr>
<tr><td>9.</td><td>Particulate matter shall be determined by either gravimetric or particle counting. Should particle counting result in failure, the gravimetric method shall be the referee.</td></tr>
<tr><td>10.</td><td>Particle counting, where equipment and laboratory capability exist, is the alternate to the gravimetric method. The limits provided below are for particulate matter for procurement requirements only. Should particle counting result in failure, the gravimetric method shall be the referee.
<table>
<tr><th>Particle counting, individual channel counts & ISO Code</th><th>Channel Counts</th><th>ISO Code 10a</th><th>Test Method
IP 565, IP 577, or D7619</th></tr>
<tr><td>≥ 4 µm (c)10b</td><td>10c</td><td>19</td><td></td></tr>
<tr><td>≥ 6 µm (c) 10b</td><td>10c</td><td>17</td><td></td></tr>
<tr><td>≥ 14 µm (c) 10b</td><td>10c</td><td>14</td><td></td></tr>
<tr><td>≥ 30 µm (c) 10b</td><td>10c</td><td>13</td><td></td></tr>
</table>
10a. Per Table 1 "Allocation of scale numbers" of ISO 4406.
10b. (c) indicates that the equipment has been calibrated in accordance with ISO 11171.
10c. Report.</td></tr>
</table>

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4. VERIFICATION

4.1 Classification of inspection. The inspection and testing of the priming fluid requirements specified herein are classified as conformance tests (see 4.2).

4.2 Conformance inspection conditions. Tests for acceptance of individual lots shall consist of tests for all requirements specified in section 3 and Table I. Conformance inspection shall include the test requirement herein.

4.2.1 Inspection lot. For acceptance purposes, individual lots shall be examined as specified herein and subject to tests for all requirements cited in section 3 and Table I.

4.3 Inspection.

4.3.1 Inspection conditions. Priming fluid supplied shall comply with the specified limiting values in Table I using the cited test methods. The specified limiting values must not be changed. This precludes any allowance for test method precision and adding or subtracting digits. For the purposes of determining conformance with the specified limiting values, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of digits used in expressing the specified limiting value, in accordance with the Rounding Method of ASTM E29.

4.4 Sampling plans.

4.4.1 Sampling. Each bulk or packaged lot of material shall be sampled for verification of product conformance in accordance with ASTM D4057 or ASTM D4177, except where individual test procedures contain specific sampling instructions.

4.4.1.1 Several priming fluid properties are very sensitive to trace contamination that can originate from sample containers. For recommended sample containers, refer to ASTM D4306.

4.4.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot and shall be subjected to the examination of filled containers as specified in 4.5.1.3.

4.5 Methods of inspection.

4.5.1 Examination of product.

4.5.1.1 Visual inspection. Samples selected in accordance with 4.4.1 shall be visually examined for compliance with 3.4.

4.5.1.2 Examination of empty containers. Before filled, each unit container shall be visually inspected for cleanliness and suitability in accordance with ASTM D4057.

4.5.1.3 Examination of filled containers. Samples taken as specified in 4.4.2 shall be examined for conformance to MIL-STD-290 with regard to packaging, packing, and marking of petroleum and related products for shipment, handling, and storage. Any container with one or more defects under the required fill shall be rejected.

4.5.2 Chemical and physical tests. Tests to determine conformance to chemical and physical properties of the propellant supplied shall be conducted in accordance with Table I. The finished propellant shall pass all tests listed in Table I. No additional testing shall be required. Requirements contained herein are not subject to corrections for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility

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tolerances may be averaged. For rounding of significant figures, ASTM E29 shall apply to all tests required by this specification.

4.5.1 Testing. Unless otherwise specified by the procuring activity, tests as specified in 3.2 shall be conducted in accordance with ASTM or IP standards and Appendix A, using applicable methods as listed in Table I.

4.6 Rejection and retest. When any sample of the priming fluid tested in accordance with 4.2 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected. Rejected material shall not be resubmitted without furnishing full particulars concerning previous rejection and measures taken to overcome defects.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The priming fluid covered by this specification is intended for use as a priming fluid in military gas turbine engines or military ramjet engines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Packaging requirements (see 5.1)
c. Quantity required and size containers required.
d. Location, injection method, and type of additives, as required.

6.3 Definitions.

6.3.1 Bulk lot. A bulk lot consists of an indefinite quantity of a homogenous mixture of material offered for acceptance in a single isolated container.

6.3.2 Packaged lot. A packaged lot consists of an indefinite number of 55-gallon drums or smaller unit packages of identical size and shape offered for acceptance and filled from the isolated tank containing a homogenous mixture of material.

6.4 Conversion of metric units. Units of measure have been converted to the International System of Units (Metric) in accordance with ASTM SI10. If test results are obtained in units other than Metric or there is a requirement to report dual units, ASTM SI10 should be used to convert the units.

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6.5 Subject term (key word) listing.

exo-Tetrahydrodicyclopentadiene
methylcyclohexane
ramjet

6.6 Safety Data Sheets. Contracting officers will identify those activities that require copies of completed Safety Data Sheets (SDS) prepared in accordance with 29 CFR 1910.1200.

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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METHOD OF TEST FOR ANALYSIS OF GRADE PRIMING FLUID (PF-1) BY GAS CHROMATOGRAPHY

A.1 SCOPE

A.1.1 Scope. This method is intended for the analysis of PF-1 priming fluid. The method provides for the quantitative determination of the two major constituents of PF-1 and for the characterization of each constituent. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 SUMMARY

A.2.1 Summary of methods. The sample is introduced into a gas chromatograph column. The detector signal is monitored continuously, and the areas of individual peaks are used for computing the quantities of the methylcyclohexane, exo-tetrahydrodicyclopentadiene, and other hydrocarbon impurities. The detailed composition of the priming fluid components is computed from the individual peak areas in the subgroup.

A.3 APPARATUS

A.3.1 Chromatograph. Any conventional gas chromatograph, having at least the following features, is acceptable.

A.3.1.1 Detector. A thermal conductivity (TCD), flame ionization (FID), or mass spectrometer (MS) detector may be used. The detector shall be operated in such a manner that its linear dynamic range is not exceeded, and its linearity should be checked periodically. The detector shall be capable of continuous operation and must be connected to the column in a manner that eliminates cold spots.

A.3.1.2 Temperature Programmer. The instrument must be capable of reproducible temperature programming over the range of 50 ºC to 300 °C. The programming rate must be sufficiently reproducible so that individual retention times do not vary by more than 2 percent. Any temperature program, including multilinear modes and intermediate periods of isothermal operation, is acceptable, provided that the column resolution, as defined in A.5.2, is not compromised. If capillary-type columns are used for the analysis, it may not be necessary to program the oven above 200 °C.

A.3.1.3 Sample inlet system. Flash vaporization will be used. For flash vaporization, the sample inlet port must be maintained between 250 °C and 300 °C, and well-conditioned septa must be used. A sample inlet splitter may be used in conjunction with capillary columns, if care is taken to ensure that a representative sample is delivered to the column. New septa should be allowed to condition overnight after being installed to minimize spurious peaks, and several blank runs should be made to purge the column of materials bleeding from the septum. Overnight conditioning can be eliminated by the practice of storing a supply of septa in the column oven. However, at least one blank run should still be made following each septum change.

A.3.1.4 Column. Open-tubular (capillary-type) columns will be used, provided the minimum resolution, specified in A.5.2, is maintained. The column should be designed for

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maximum thermal stability and minimum bleed since the temperature program is likely to run as high as 275 °C for packed columns and 200 °C for capillary columns. Dual-column compensation is required if column bleed results in excessive baseline drift at high temperature.

A.3.1.5 Data Acquisition. Computer controlled data acquisition and integration is the preferred method of data collection and analysis. The computer system must be designed and calibrated for gas chromatograph operation.

A.3.1.6 Microsyringe. A microsyringe, capable of delivering sample volumes from 0.5 to 5.0 microliters, is required for FID and TCD detectors. The widely used 0 – 10 microliter syringe, Hamilton Model 701N or its equivalent, is recommended. A microsyringe, capable of delivering sample volumes from 0.1 to 0.5 microliters, is required for MS detectors. These syringes are routinely available from most instrument manufacturers and chromatography supply houses.

A.4 REAGENTS AND MATERIALS

A.4.1 Stationary phase. The stationary phase should be one of the familiar non-polar polysiloxanes, or equivalent. The following columns have been used successfully:

100% Dimethylpolysiloxane

(5%-Phenyl)-methylpolysiloxane

Either of the listed phases is satisfactory, provided baseline drift at high temperatures is maintained by conditioning or by dual-column compensation.

A.4.2 Carrier gas. Either helium or nitrogen is suitable for FID detectors, although nitrogen is known to give somewhat higher column efficiencies at lower flow rates and is more economical. Helium must be used with thermal conductivity detectors. Either helium or hydrogen may be used with a mass spectrometer detector. The flow must be maintained constant to within one percent throughout the entire range of operating temperatures by means of constant flow controllers.

A.4.3 Hydrogen. High purity, for flame ionization detectors.

A.4.4 Air. Dry, filtered, and oil-free, for flame ionization detectors.

A.4.5 Exo-tetrahydrodicyclopentadiene. Minimum ninety-four weight percent purity, hereafter referred to as C-10.

A.4.6 Methylcyclohexane. Minimum ninety-nine weight percent purity, hereafter referred to as MCH.

A.4.7 N-pentane, n-hexane, or petroleum ether. Chromatographic purity.

A.4.8 Qualitative calibration mixtures. A series of known hydrocarbons is used for calculating column resolution. Prepare a mixture containing about one percent (by volume) of MCH, n-dodecane, and n-tetradecane in a suitable solvent such as pentane, hexane or petroleum ether. This mixture will be used to compute the column resolution.

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A.5 PREPARATION OF APPARATUS

A.5.1 Column. Any conventional method of conditioning and installing the column is satisfactory, provided the system is free of gas leaks and the column resolution meets the minimum requirement specified in A.5.2.

A.5.2 Column resolution. Run a sample of the qualitative calibration mixture and measure the retention times and widths of the n-dodecane and n-tetradecane peaks as shown on Figure A-1. Resolution is defined as the difference in retention times for n-C_{12} and n-C_{14} divided by their average width:

$$R = \frac{2\left(RT_{14} - RT_{12}\right)}{W_{14} + W_{12}}$$

Resolution determined in this way must be at least 5.0 and can be optimized by varying the temperature program and the carrier flow rate. If resolution cannot be increased to at least 5.0 by optimizing these parameters, it is necessary to use a different column.

A.5.3 Chromatograph. Place in service according to the manufacturer's specifications paying particular attention to flow rates through the detector.

A.6 PROCEDURE

A.6.1 Suitable sample volume. Establish a suitable sample volume by making several trial runs with the qualitative calibration mixture. For FID instruments, the volume may be between 0.5 and 1.0 µL. For TCD instruments, volumes from 1.0 to 5.0 µL have been found satisfactory. For MS instruments, volumes from 0.1 to 0.25 µL have been found satisfactory.

A.6.2 Sample. Shake and/or mix the neat PF-1 sample. If the operator dilutes the sample with a solvent, account for the dilution in the final calculation. Prepare an aliquot for injection into the GC system. Inject a measured volume of sample into the GC system and activate the temperature program, data recorder, and/or integrator.

A.6.3 Integration. Integrate the resulting chromatograph for all the peaks. With either the MCH or C-10 peak, when there are resolved peaks within plus or minus 10 percent of the retention time, add the areas under those peaks into the area of associated peak provided the area does not contribute more than 20 percent to that peak. (Area under J-10 peak = $A_{J\text{-}10}$; Area under MCH peak = A_{MCH}; Area under other peaks = $A_{impurities}$ in A.7.1.) A typical GC/FID chromatogram is included as Figure A-2.

A.6.3.1 Methylcyclohexane. If commercial MCH was used to blend the PF-1 acceptance, appreciable amounts of heptane and toluene may appear as resolved peaks near the MCH peak. These peaks should be added, uncorrected, to the area of the MCH peak. Assuming relative response factors for these materials does not seriously affect the accuracy of the method.

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A.7 CALCULATIONS

A.7.1 Weight percent. For each PF-1 acceptance sample, calculate the weight percent of MCH and C-10 as follows:

$$\text{wt\% C-10} = \frac{A_{C\text{-}10}}{A_{C\text{-}10} + A_{MCH} + A_{impurities}} \times 100$$

$$\text{wt\% MCH} = \frac{A_{MCH}}{A_{C\text{-}10} + A_{MCH} + A_{impurities}} \times 100$$

$$\text{wt\% impurities} = 100 - (\text{wt\% C-10} + \text{wt\% MCH})$$

In the above equation, all hydrocarbons other than those contained in the MCH and C-10 subgroups are considered impurities.

A.8 PRECISION

A.8.1 Studies. Detailed precision studies have not been performed for this method. However, limited results indicate that the method is capable of the following precision.

A.8.1.1 Repeatability. Replicate analyses, performed under identical conditions in the same laboratory, should yield results that do not vary more than 1 percent (absolute) from the mean value for each component.

A.8.1.2 Reproducibility. When the same sample is analyzed in different laboratories according to this method, the results for any component should not differ by more than 1.5 percent (absolute).

A.8.1.3 Accuracy. Back-analysis of a calibration standard should give results that agree with the known values to within 0.5 percent (absolute).

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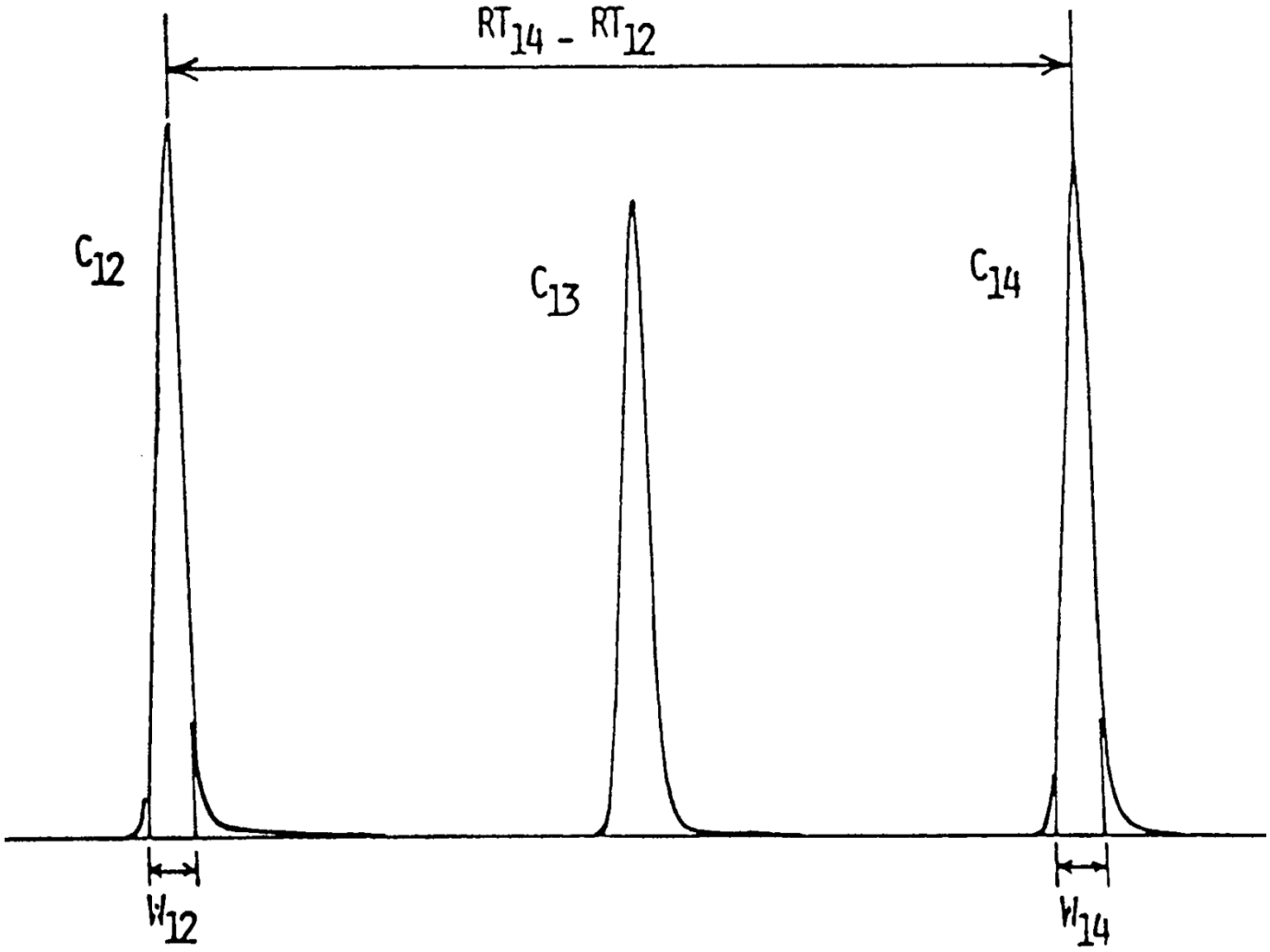
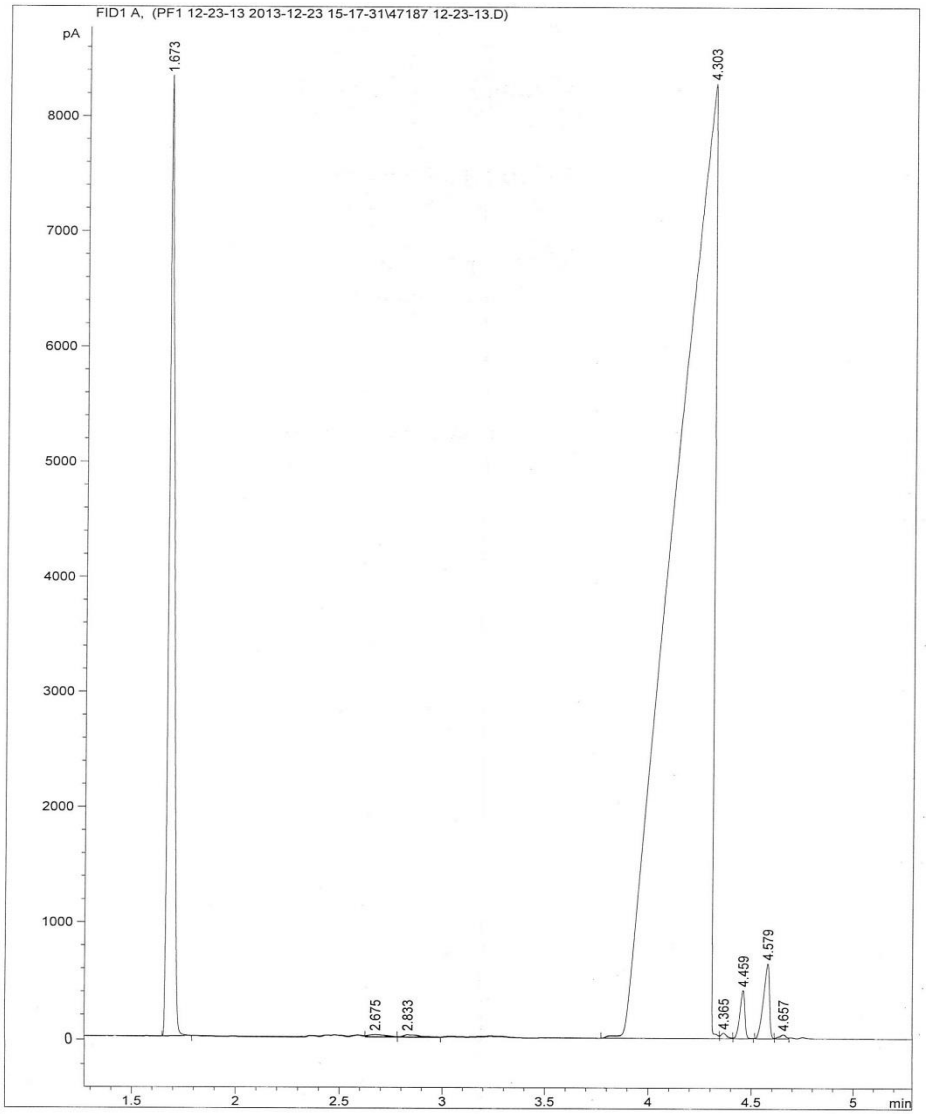


FIGURE A-1. Calculation of column resolution.

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Peak #	R.T. [min]	Type	Width [min]	Area [pA*s]	Height [pA]	Area [%]	Identification
1	1.673	BB	0.0240	1.48150E+04	8327.3975	11.7926	MCH
2	2.675	VV	0.0720	75.1073	13.2754	0.0598	Impurities
3	2.833	VV	0.0659	61.7312	12.2622	0.0491	
4	4.303	BV	0.1554	1.08599E+05	8279.8340	86.4436	C-10 and associated peaks
5	4.365	VV	0.0283	73.1312	36.8774	0.0582	
6	4.459	VB	0.0247	666.3780	407.5647	0.5304	
7	4.579	BV	0.0298	1279.4342	637.2098	1.0184	
8	4.657	VV	0.0342	60.1546	24.8574	0.0479	
		Totals:		1.25630E+05	1.77393E+04		

FIGURE A-2. Typical GC/FID chromatogram of PF-1.

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CONCLUDING MATERIAL

Custodians:
Navy – AS
Air Force – 68
DLA – PS

Preparing activity:
Air Force - 68
(Project 9135-2024-007)

Review activities:
Air Force – 11

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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